

Stress Analysis of Boiling water Reactor with $r/t = 10$ using Ansys & Lame's Equation

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Introduction to Thick and Thin Cylinders

Definition

A cylinder is classified as **thick** when the ratio of inner radius to thickness less than 10, such that radial stress cannot be neglected and hoop stress varies significantly across the wall thickness.

Common Applications

- Hydraulic cylinders
- Gun barrels
- High-pressure pipes
- Pressure vessels

Key Distinction

Thin Cylinder

Uniform hoop stress across thickness. Radial stress negligible.

$$r/t > 10$$

Thick Cylinder

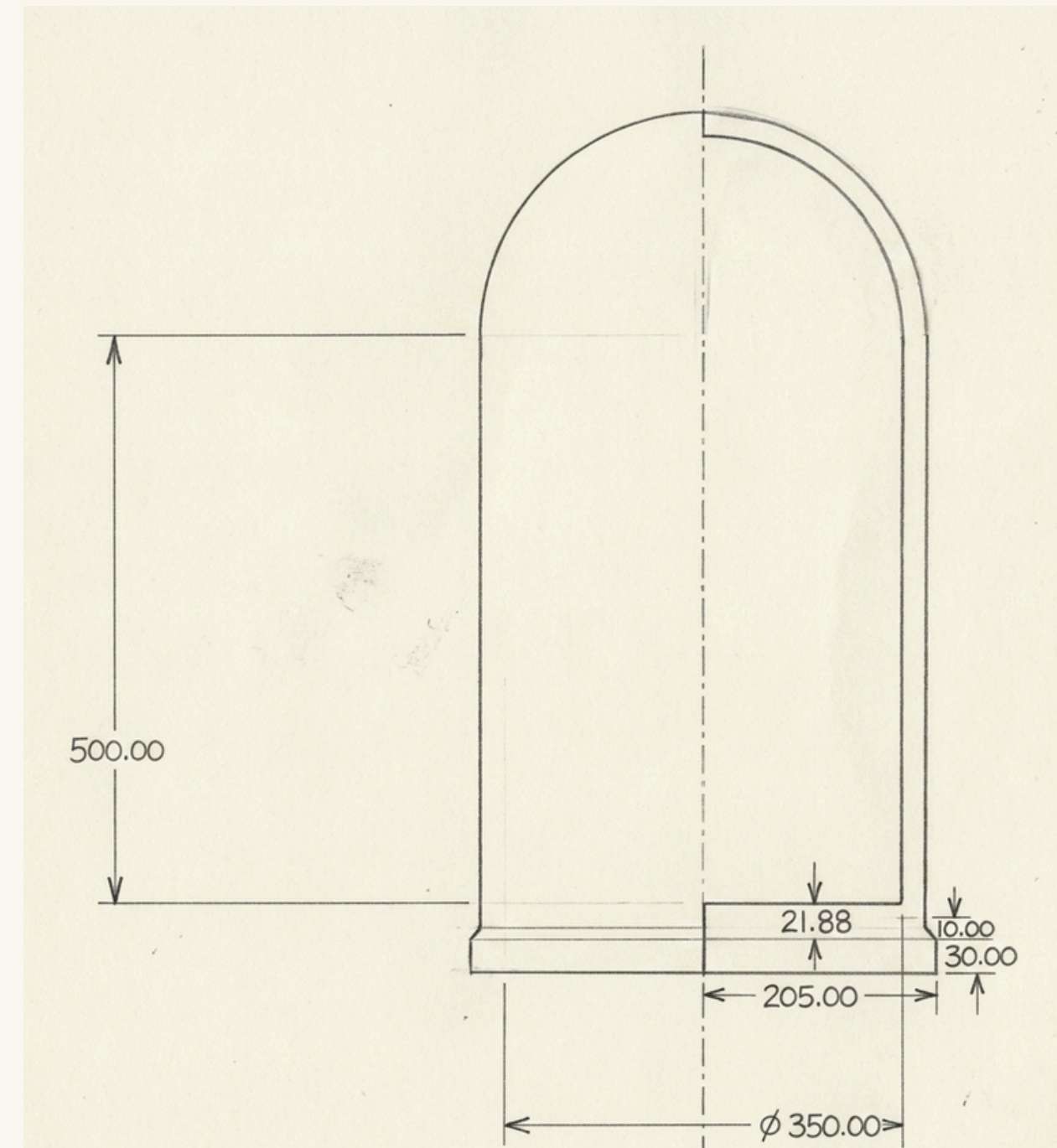
Stress varies significantly across wall. Radial stress substantial.

$$r/t < 10$$



Problem Statement

To compare radial, Longitudinal and hoop stress distribution in a thick and thin cylindrical and hemispherical reactor at $r/t = 10$ and validate analytical results using Ansys simulation.





Theoretical Calculation - Thick

Lame's Equation —

$$\sigma_r = A - \frac{B}{r^2} \quad \text{--- (I)}$$

$$\sigma_\theta = A + \frac{B}{r^2} \quad \text{--- (II)}$$

Boundary Conditions —

$$\text{At } r = R_1 ; \quad \sigma_r = -P_i$$

$$\text{At } r = R_2 ; \quad \sigma_r = 0$$

Putting boundary conditions in eq. (I) & (II)

$$-P_i = A - \frac{B}{R_1^2} \quad \text{--- (A)}$$

$$\& \quad 0 = A - \frac{B}{R_2^2} \quad \text{--- (B)}$$

from eq. (B),

$$A = \frac{B}{R_2^2}$$

$$\Rightarrow -P_i = \frac{B}{R_2^2} - \frac{B}{R_1^2}$$

$$\Rightarrow B = \frac{-P_i R_1^2 R_2^2}{R_1^2 - R_2^2} = \frac{P_i R_1^2 R_2^2}{R_2^2 - R_1^2} \quad \text{--- (X)}$$

$$\Rightarrow A = \frac{P_i R_1^2}{R_2^2 - R_1^2} \quad \text{--- (Y)}$$



Theoretical Calculation - Thick

Formulas for Radial & Hoop stress at a point

$$\sigma_r = \frac{P_i R_i^2}{R_o^2 - R_i^2} \left(1 - \frac{R_o^2}{r^2} \right)$$

$$\sigma_\theta = \frac{P_i R_i^2}{R_o^2 - R_i^2} \left(1 + \frac{R_o^2}{r^2} \right)$$

$$\sigma_L = \frac{P_i R_i^2}{R_o^2 - R_i^2}$$

Formulas for Stresses at Hemispherical Ends

$$\sigma_h = \sigma_L$$

$$\sigma_h = \frac{P_i (2 R_i^3 + R_o^3)}{2 (R_o^3 - R_i^3)}$$



Theoretical Calculation- Thick

As we know,

$$\text{Radius Inner } (R_i) = 175 \text{ mm}$$

$$\text{Thickness } (t) = 21.875 \text{ mm}$$

$$\text{Outer Radius } (R_o) = 196.875 \text{ mm}$$

$$\text{Length of cylindrical part} = 500 \text{ mm}$$

By calculating stress at cylindrical & hemispherical ends at various radius, we made this table.

Theoretical Results - Thick Cylindrical End

Location	Radial Stress (MPa)	Hoop Stress (MPa)	Longitudinal Stress
Inner Radius (175 mm)	-7.5	63.97	28.23
Outer Radius (196.8mm)	0	56.47	28.23
Interface (186 mm)	-3.39	59.86	28.23



Theoretical Results - Thick Hemispherical End

for Hemispherical End —

$$\sigma_h = \frac{P_i (2R_i^3 + R_o^3)}{2(R_o^3 - R_i^3)}$$

As $\sigma_h = \sigma_l$; $P_i = 7.5 \text{ MPa}$; $R_i = 175 \text{ mm}$; $R_o = 196.87$

$$\sigma_h = \sigma_l = \frac{7.5 \times (2 \times (0.175)^3 + (0.196875)^3)}{2((0.196875)^3 - (0.175)^3)}$$

$$\sigma_h = 30.29 \text{ MPa}$$



Theoretical Results - Thin Cylindrical and Hemispherical End

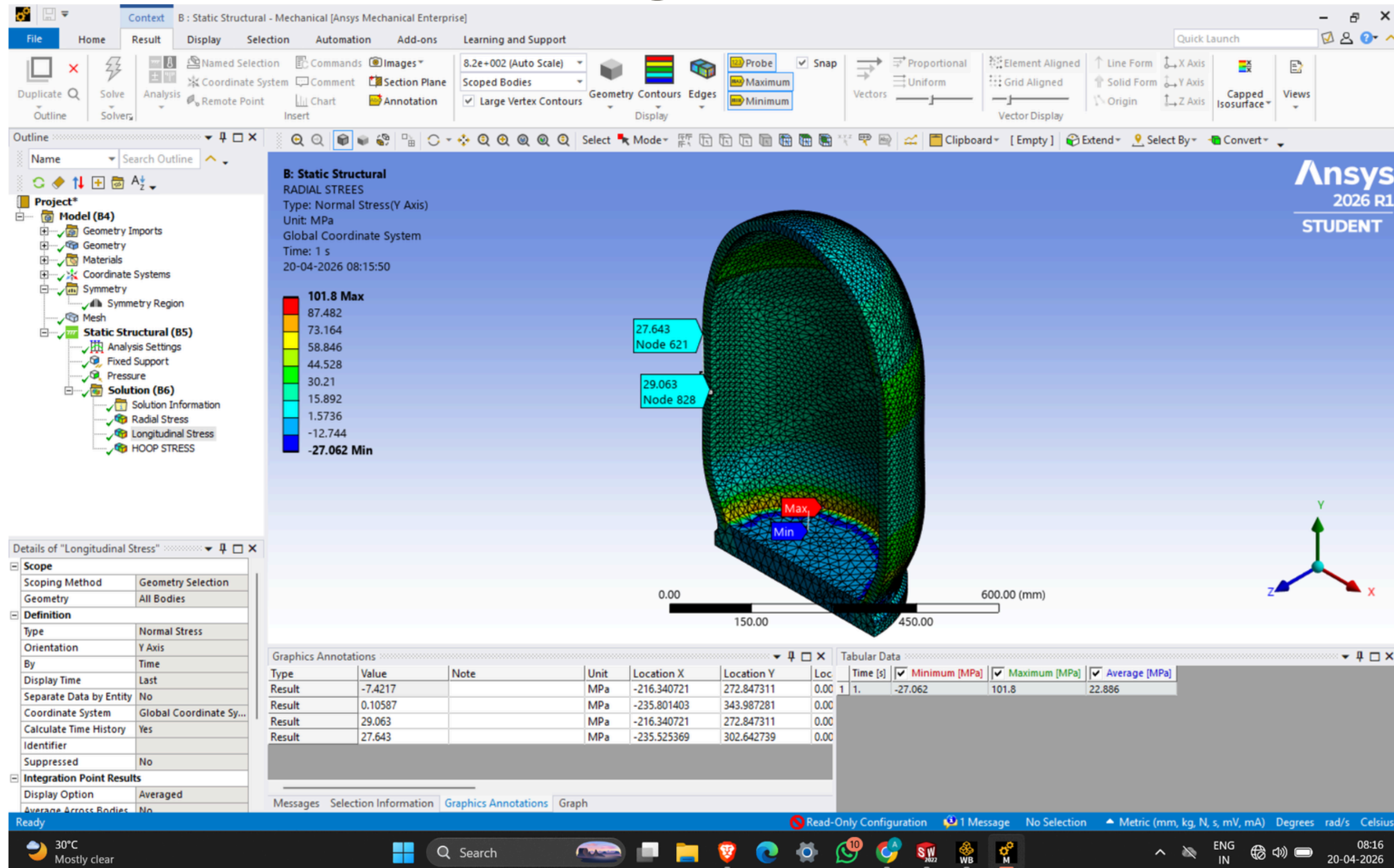
for thin cylindrical & Hemispherical End —
Radial Stress is negligible, Therefore

$$\text{Hoop stress} = \frac{P_i R_i}{t} = \frac{17.5 \times 175}{21.875} = 60 \text{ MPa}$$

$$\text{Longitudinal stress} = \frac{P_i R_i}{2t} = \frac{7.5 \times 175}{2 \times 21.875} = 30 \text{ MPa}$$



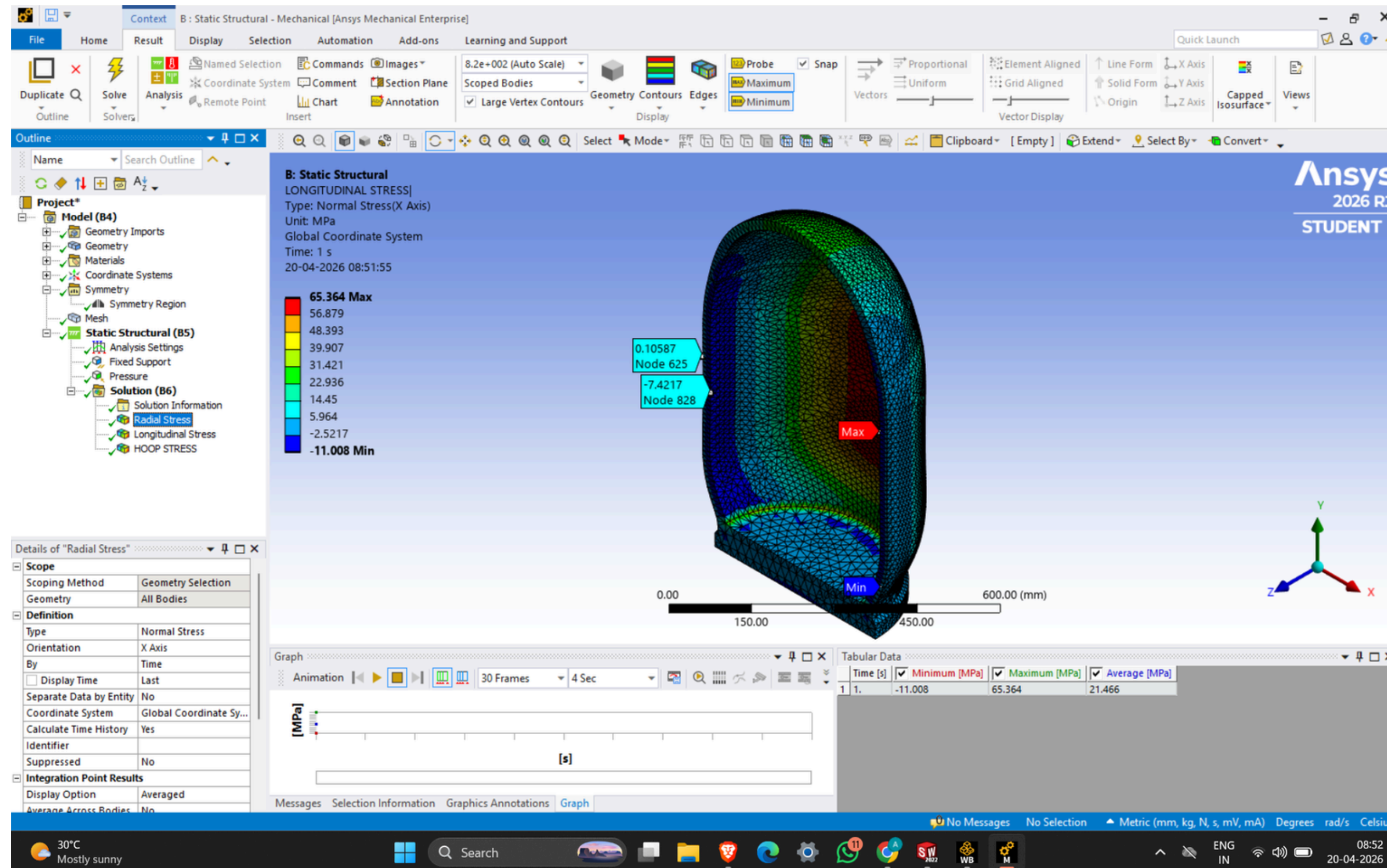
Simulation Result : Longitudinal Stress



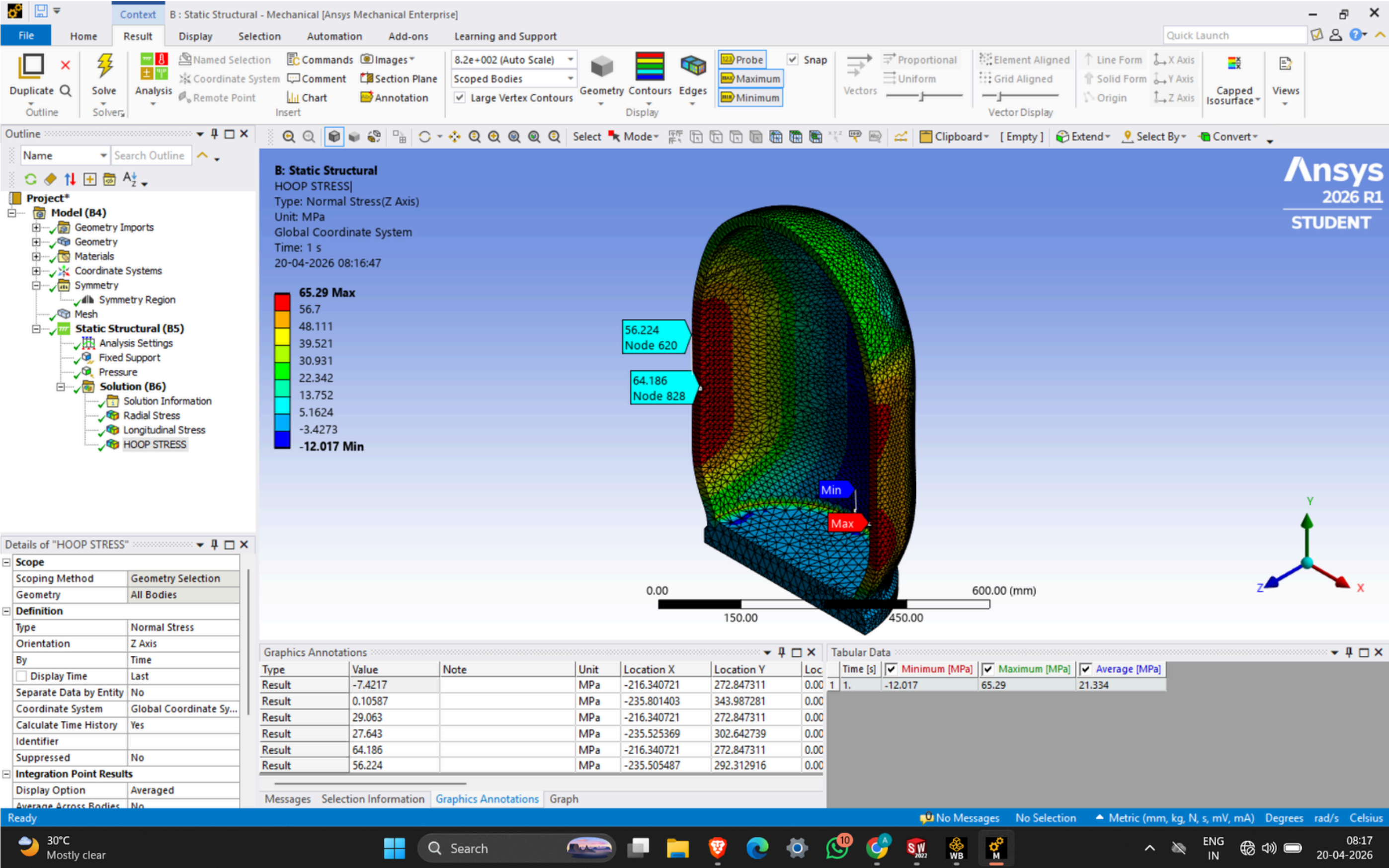
Simulation Result : Radial Stress



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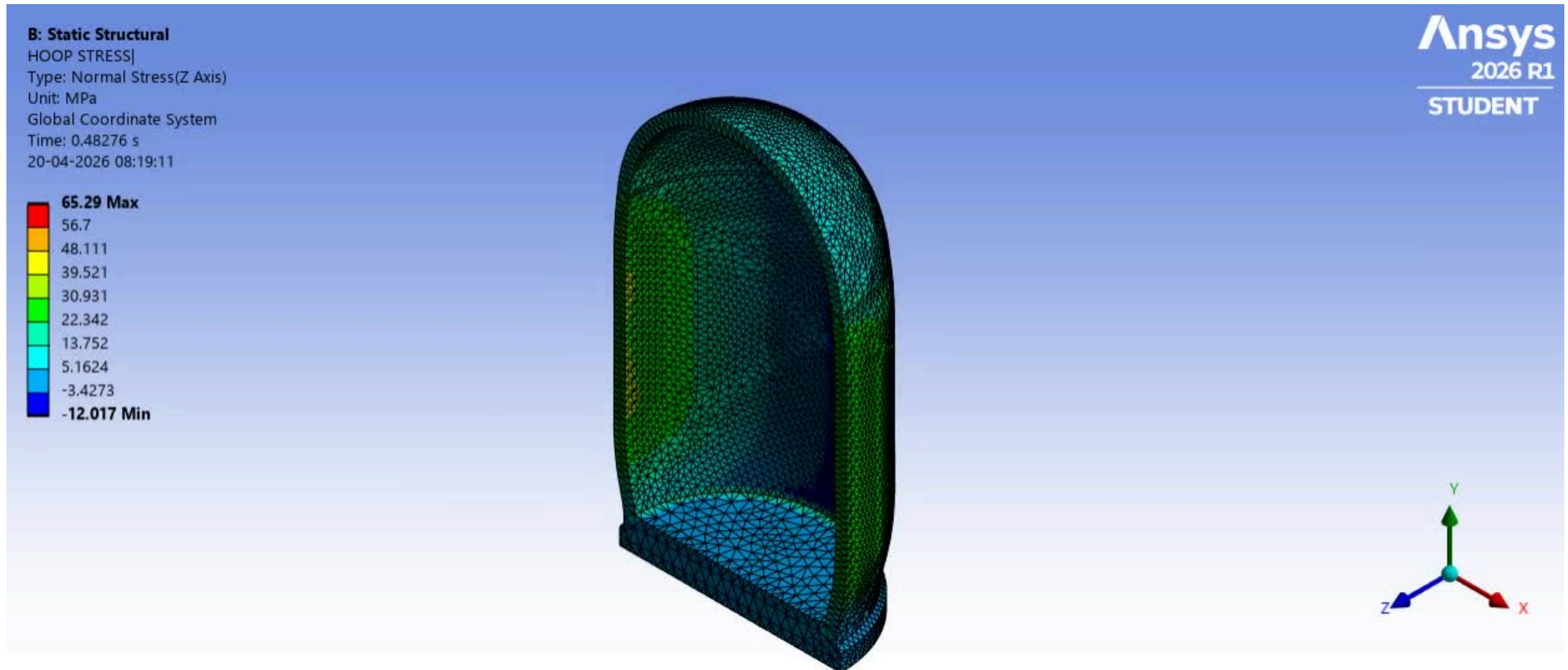
Simulation Result : Hoop Stress



Simulation Result : Video



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Simulation Result - Cylindrical

Location	Radial Stress (MPa)	Hoop Stress (MPa)	Longitudinal Stress
Inner Radius (175 mm)	-7.42	64.18	29.06
Outer Radius (196.875 mm)	-0.1	56.22	27.64
Interface (186 mm)	-3.59	60.18	28.31

Simulation Result - Hemispherical

Location	Radial Stress (MPa)	Hoop Stress (MPa)	Longitudinal Stress
Inner Radius (175 mm)	-7	30.225	30.1
Outer Radius (196.875 mm)	-0.19	26.6	26.64
Interface (186 mm)	-3.52	28.44	28.44

Percentage Error (%) - Cylindrical

Location	Radial Stress (%)	Hoop Stress (%)	Longitudinal Stress(%)
Inner Radius(175 mm)	-1.07	0.33	2.94
Outer Radius (196.8 mm)	approx. 0	-0.44	-2.09
Interface (186 mm)	5.9	0.53	0.28



Percentage Error (%) - Hemispherical

for Hemispherical End,

Theoretical = 30.29 MPa

Simulation = 30.1 MPa

$$\text{Error} = -0.63\%$$

Thin v/s Thick

Feature	Thin Cylinder	Thick cylinder
Condition	$r/t > 10$	$r/t < 10$
Hoop Stress	Uniform	varies across thickness
Radial Stress	Negligible	Significant
Accuracy	Approximate	More accurate

Conclusion

- Stress distribution in thick cylinders is non-uniform across thickness, unlike thin cylinder assumption
- Hoop stress is maximum at inner surface and decreases outward
- Radial stress is significant and cannot be neglected in thick cylinders
- Simulation results show excellent agreement (within ~1–5%) with Lamé's theory
- Higher error in radial stress is due to mesh sensitivity and size
- Thin cylinder theory underestimates hoop stress and ignores radial stress
- Hemispherical end shows nearly uniform stress distribution, validating design advantage
- The study confirms that Lamé's equation accurately predicts real stress behavior



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Thank You